CONTINUED

You will be asked to calculate the size of a sound file to a certain unit of measurement. The answer given (10 584 000) is the number of bits that are in the sound file. If you wanted to know how many bytes are in the sound file, you need to divide the final answer again by 8, as there are 8 bits in each byte, for example:

$$10\ 584\ 000 \div 8 = 1\ 323\ 000 \text{ bytes}$$

If you wanted to know how many mebibytes the file was, you need to divide the answer in bytes by 1024, then by 1024 again, for example:

$$1\ 323\ 000 \div 1024 \div 1024 = 1.3 \text{ MiB (to 1 decimal place)}$$

1.5 Data compression

You may already have experience of having large files and struggling to have enough storage space to store them. One thing that can be done to reduce the size of a file is **compression**.

Compressing a file will reduce the size of the file that can have several benefits, such as:

- Not as much storage space is needed to store the file.
- It will take less time to transmit the file from one device to another.
- It will be quicker to upload and download the file.
- Not as much bandwidth is needed to transmit the file over the internet.

There two type of compression that can be used, **lossy** and **lossless**. Each compresses the data in a different way.

Lossy compression uses a compression algorithm that finds the unnecessary and redundant data in the file. This data is permanently removed from the file. This type of compression is mainly used on an image file or a sound file.

Unnecessary data in an image file that could be removed are colours that the human eye cannot distinguish. This means that a lossy compression algorithm for an image file could reduce the size of the file by reducing the colour depth. It could also reduce the size of the file by reducing the image resolution, reducing the number of pixels used to create the image.

Unnecessary data in a sound file that could be removed are sounds that cannot be heard by the human ear. It could also be the removal of softer sounds that are played at the same time as louder sounds, as humans will only hear the louder sound. The method that is used to find these kinds of sounds in a sound file is called **perceptual music shaping**. This is the most common type of method that is used in the file format MP3. The size of a sound file can also be reduced by reducing the sample rate and the sample resolution.

KEY WORDS

compression: a method that uses an algorithm to reduce the size of a file.

lossy: a compression method that reduces the size of a file by permanently removing data.

lossless: a compression method that reduces the size of a file by temporarily altering the data.

KEY WORD

perceptual music shaping: a process that is used in lossy compression that removes sounds that are not normally heard by the human ear.

Questions

7 Why do you think lossy compression is not commonly used on text files?

8 Will lossy compression affect the quality of an image? Why?

9 Will lossy compression affect the quality of a sound recording? Why?

SELF ASSESSMENT

What were the points that you thought about regarding lossy compression to answer question 7? Did you relate these points back to why lossy compression is not suitable for a text file, or did you just make points about the drawbacks of lossy compression in general?

Lossless compression uses a compression algorithm to reduce the size of the file, without permanently removing any data. A common method that can be used to do this is by finding repeating data in the file and grouping this data together. For example, consider the following message:

WHEN IT IS SNOWING HEAVILY LOOK OUTSIDE.
LOOK OUTSIDE IT IS SNOWING HEAVILY.

Excluding the spaces between the words and the full stops, the message has a total of 62 characters. 1 character requires 1 byte of storage, so 62 bytes of storage would be needed to store this message.

When you look at the message, you should notice that it consists of words that are mostly repeated. Instead of storing all 62 characters individually, the words and the positions at which they occur in the message can be stored in a lookup table, for example:

Word	Position(s) in the message
WHEN	1
IT	2, 10
IS	3, 11
SNOWING	4, 12
HEAVILY	5, 13
LOOK	6, 8
OUTSIDE	7, 9

The lookup table needs 1 byte for each character in each word and 1 byte for each position the word occurs in the message. Therefore, 33 bytes are needed to store the words and 13 bytes to store the positions, giving a total of 46 bytes. This is much less than the 62 bytes we required with our original method. No data has been lost and we have reduced our storage requirements by 26%, quite a saving! To recreate the message, the computer simply retrieves the words and places them in the positions allocated.

A similar process can be applied to an image file with a lossless compression algorithm called **run length encoding (RLE)**. This algorithm will group together repeating pixels and store how many times they occur. For example, consider the simple image in Figure 1.6:

KEY WORD

run length encoding (RLE): An algorithm that groups together repeating patterns and indexes them.

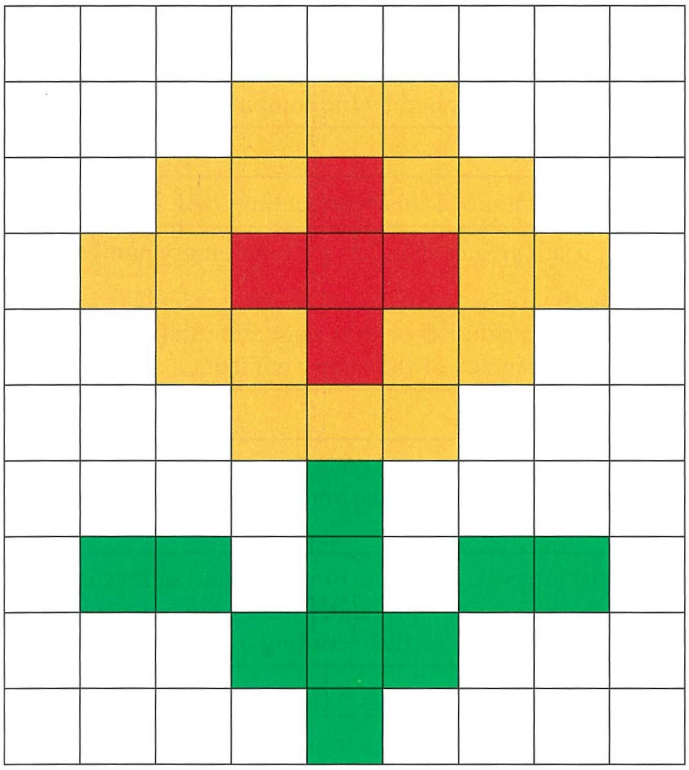


Figure 1.6: A simple pixel image

If W is white, Y is yellow, R is red and G is green, RLE could compress the image into the following data:

12W, 3Y, 5W, 2Y, 1R, 2Y, 3W, 2Y, 3R, 2Y, 3W, 2Y, 1R, 2Y, 5W, 3Y, 7W, 1G, 5W, 2G, 1W, 1G, 1W, 2G, 4W, 3G, 7W, 1G, 4W.

This means that the amount of data can be reduced as pixels are grouped together and the data for each pixel does not need to be stored. The resolution of the image is stored with the data, so that the computer knows the size to recreate it when the data is decompressed.

ACTIVITY 1.4

Why don't you try creating a simple image and writing out the RLE data for it? You could then give it to your friend, along with the image resolution, and they can have fun trying to recreate your image.

Peer Assessment

Ask your friend whether you got the RLE correct for your image in Activity 1.4. Ask them if you could have done anything to make your RLE any clearer. Ask them if you missed any important data.